

C11BA

Business Analytics and Introduction to Big Data

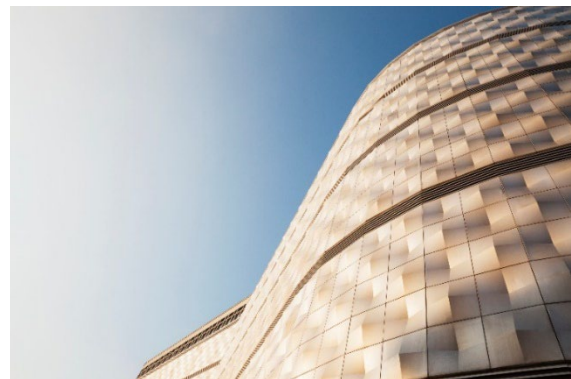
COURSEWORK – CASE STUDY ANALYSIS

A case developed by Dr Fabienne Chedid

DLS Mall - a problem worth solving

THE COMPANY AND BACKGROUND

DLS Group is a family-owned, leading retail corporation that operates a range of shopping malls and department stores across Dubai. The primary aim of the business is to offer a wide variety of products and services to cater to the diverse needs of both residents and tourists. The group's flagship shopping mall, **DLS Mall**, is one of Dubai's premier shopping destinations, known for its luxury brands, diverse food courts, and entertainment options. **DLS Mall** has a vast retail space that hosts a mix of high-end boutiques, electronics stores, fashion outlets, and specialty shops. As a key player in the retail industry in Dubai, **DLS Group** has developed its expertise in managing a broad range of department categories, from fashion and beauty to technology and home goods, aiming to deliver exceptional customer experiences.



In response to the impact of COVID-19 on sales, **DLS Mall's** management is re-evaluating how product categories, displays, and inventory are managed to boost efficiency and profitability. With a focus on understanding post-pandemic sales distribution across the mall, they have engaged a consultancy firm to conduct a scoping study and identify opportunities for improvement, all while ensuring that customer satisfaction remains the priority.

Product Categories

DLS Mall offers a vast array of categories, including high-end fashion, electronics, home decor, gourmet foods, and beauty products. The focus of this case study is on the effective



management and display of these categories within the mall's retail environment. With the rapid growth in customer demands and the ever-changing trends in retail, **DLS Mall** has expanded its product offerings over time, leading to a complex inventory that includes items ranging from luxury handbags and designer clothing to the latest gadgets and gourmet treats.

To support this expansive inventory, **DLS** Mall aims to allocate resources such as salespeople, merchandisers, and procurement specialists as efficiently as possible. These roles are crucial in ensuring that products are not only sourced efficiently but also displayed in a manner that maximizes consumer appeal and sales performance. Also, the allocation of these resources should be carefully managed to meet the dynamic needs of the market, ensuring that each product category is adequately supported and that customer experiences within the Mall are optimized.

Inventory Management

DLS Mall offers a wide variety of product categories, each designed to cater to the diverse preferences of its customers. The management team has noticed that most of the sales and inventory space is focused on just a few key categories, while many smaller categories bring in only a small portion of sales. The management team would like to know which specific categories these are and where they are located. This necessitates careful management to avoid issues such as overstocking in less popular categories and understocking in high-demand ones, ultimately ensuring a balanced and efficient product offering across the mall.

DLS Mall manages its inventory and product distribution through a central warehouse and several smaller backup facilities within the mall. Products are distributed to various stores within the mall based on sales forecasts and replenishment plans. However, the current system has led to challenges in resource allocation, inventory management, and display management. For instance, there have been instances where newly introduced products were over-ordered, leading to excess stock that occupies warehouse space and valuable retail space, while other popular items were understocked, resulting in lost sales opportunities and poor customer satisfaction.

Customer Interaction and Sales

Customers visiting **DLS** Mall have different floors and areas within the mall where they can shop—such as the West wing, East wing, Ground Floor, First Floor, etc., which play a critical role in the overall shopping experience and impact the management of displays and inventory. For instance, popular items at the main entrance may sell out quickly, while similar products in less frequented areas remain stocked. Likewise, popular categories displayed prominently at the main exit may attract quick sales, while similar categories in less frequented areas may sell slower due to lower visibility.

A TIME FOR CHANGE

The management team at **DLS** Mall feels that now is the time to reassess how the business manages its product categories, display and inventory to improve efficiency and profitability. They believe that there is an opportunity to streamline operations by focusing on key product categories and key regions within the Mall that contribute the most to the Mall's revenue. However, there is a strong culture within **DLS** Group that emphasizes customer satisfaction above all else, and any changes must not negatively impact the customer experience. Without the time available or the key skills necessary to assess the problem, identify solutions and implement the changes needed, **DLS** Mall has engaged your consultancy firm to assist in this transformation. Following initial discussions with **DLS** Mall managers, the consultancy firm has been asked to conduct a scoping study to better understand the current situation and identify potential opportunities for improvement.

Team Briefing

As the consultancy firm's Business Analyst assigned to the **DLS** Mall project, your supervisor has asked you to take a first look at the data sets provided by the Mall's sales and inventory planners. Your supervisor wants to understand the 'size and shape' of the various categories, the different locations, floors and areas within the mall. Initial discussions suggest that there may be an opportunity to rationalize both the category range and area distribution, but first, the scope of that opportunity needs to be quantified and explored to get a better understanding of the problem. Additionally, an analysis of customer spending patterns is needed to determine which segments contribute most to the Mall's revenue and which are less profitable.

Keen to impress your boss, you are ready to tackle the challenge and begin planning the necessary analyses, keeping in mind that the data sets provided may be messy and may require reasonable assumptions. After all, this is the reality of working with real business data!

Your goal is to prepare a report that provides the insights that will guide your supervisor in the decision-making process, helping her decide on the next steps in the project.

Your Report

Produce a concise report of no more than 1200 words clearly presenting your findings to your employer (NOT **DLS** Mall). Please refer to the Report template example which you may wish to use.

Your report should include:

- **Title page:** Your company name, project title, prepared by you, date, (**see template**)
- **Executive Summary:** A short summary of the whole report including key findings and next steps.
- **Introduction:** A brief introduction clearly telling the reader (your boss) the purpose of the report and what is included in the report
- **Background:** A brief background to provide context and to demonstrate your understanding of the overall project
- **Methodology:** Explain your approach to the task, what methods, tools, techniques you used to analyse the data and WHY. What data did you use.
- **Results:** Present and discuss your results. Use appropriate graphical and/or tabular output that best illustrates what you have uncovered in the data. Each figure and table must be discussed, explaining the relevance and importance of the findings. Ensure axis labels are correct.
- **Conclusions:** Clearly state what can be concluded from the analyses you have carried out. Each conclusion must be substantiated.
- **Next steps:** What do you suggest should happen next? What further analyses do you recommend and why? What additional data should the team request from DLS Mall?

Further guidance

- The maximum word count (1200) does NOT include Title page, Executive Summary, tables, figures, appendices or reference list.
- The Executive Summary must be less than one side of A4.
- Style /Font: Calibri or Arial, 11 or 12 pt, 1.2 line spacing.
- Tables and Figures must be professionally presented. Do not use dark or coloured backgrounds.
- Any formulae must be presented using Equation Editor (available in Microsoft Word)
- You should reference a small number of very relevant sources if they support and help to justify the approach/methods you adopt /results you obtain. Use the Harvard referencing method.

Submit via Canvas on Tuesday 22 October 2024 before 12:00 (UK time) / 15:00 (Dubai time).